COMP 248 / GROUP 1 / HEALTH INQUIRIES

Sleep Health Research

Multi-agent design and prototype report

AI Systems Design | Course shell section 401

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Scope and rationale

Research Analyst workflow for general sleep-health questions. The prototype indexes four curated paraphrases of CDC About Sleep, covering duration, quality, habits, and benefits. The scope gate is a keyword heuristic and is evaluated as a routing rule. Use general example questions without personal details.

Assumptions: English questions, one curated CDC page, a small fixed corpus, local execution, and a two-worker prototype. The source is United States public-health material and its coverage varies across populations.

Architecture and UML components

Architecture: Research Analyst > Streamlit > LangGraph planner/router > research worker > reflective worker. The research worker has access to the RAG tool. The reviewer receives only the draft and retrieved passages through graph state.

UML component dependencies:

[Streamlit] --> [LangGraph router]

[LangGraph router] --> [Research worker]

[LangGraph router] --> [Reflective worker]

[Research worker] --> [LangChain retrieval tool] --> [ChromaDB]

Each arrow denotes a dependency. The router selects a worker path and coordinates state. Chroma stores chunk ID, section, paraphrase, source URL and binary term vector. The research worker reads the index through the tool. The reviewer reads a snapshot of retrieved records from state.

Workflows and UML interaction

Router: question > sleep scope and personal-advice check > research or scope reply.

Research worker: query > invoke RAG tool > retrieve up to three matching chunks > assemble cited draft.

Reflective worker: receive draft and evidence > compare every citation ID and statement > publish, abstain, or flag unsupported output.

Sequence, top to bottom:

1. Analyst -> Streamlit: submit question

2. Streamlit -> LangGraph router: invoke workflow

3. Router -> Research worker: route in-scope question

4. Research worker -> RAG tool -> ChromaDB: query by vector

5. ChromaDB --> RAG tool --> Research worker: ranked passages

6. Research worker -> Reflective worker: draft and evidence

7. Reflective worker --> Streamlit: reviewed summary or reason

8. Streamlit --> Analyst: display result and applicable source links

Alternatives: out-of-scope input exits at step 3. Empty retrieval produces an insufficient-evidence verdict. A mismatched draft is withheld.

Design decisions and responsible AI

Routing: a flat rule set is readable and repeatable for one topic. A model router could generalize to paraphrases, at the cost of additional evaluation, latency, and a provider dependency.

RAG placement: an explicit LangChain tool gives the research worker a clear invocation boundary. Retrieval integrated into the storage service could centralize access but would hide tool invocation from the worker design.

Drafting: the research worker composes a short extractive summary from curated paraphrases. A generative LLM could synthesize across a larger collection and adapt its answer more closely, with more complex evidence verification. The worker uses no LLM in this prototype.

Reflection: exact matching checks citation IDs and text deterministically. A model-based reviewer could evaluate paraphrases but would need human calibration. The reviewer uses no LLM and checks grounding at the passage level. The deterministic router also uses no LLM.

Privacy: use public documents and avoid personal information in prompts. The app does not write prompts to persistent storage. Fairness: test coverage across age groups and wording variants. Explainability: expose source links and review verdicts. Responsibility: review CDC source freshness and keep output within the indexed scope.

Evaluation and demo plan

Use labeled questions and expected relevant chunks. Precision@3 measures the fraction of returned chunks that are relevant, with an explicit top-three evaluation convention. Recall@3 measures the fraction of all relevant chunks found in the top three. Track route accuracy across supported and unsupported queries.

Measure supported-claim rate by human labeling of answer claims against their passages. Track reviewer false acceptance on altered drafts. Exact matching is appropriate for this extractive prototype and cannot assess nuanced paraphrases or medical correctness.

Demo: ask about sleep habits, then ask about sleep on Mars for insufficient evidence. Use the test suite to exercise a mismatched draft and a personal-medication route.

Conclusion: the two-worker flow exposes evidence and review outcomes. Before expanding, measure retrieval against a labeled set and test paraphrases and age-group coverage.

References and submission records

Centers for Disease Control and Prevention. About Sleep. https://www.cdc.gov/sleep/about/index.html

LangChain tool reference. https://docs.langchain.com/oss/python/langchain/tools

LangGraph graph API. https://docs.langchain.com/oss/python/langgraph/graph-api

COMP 248 Project Assignment V2, course handout.

AI disclosure: Yes. Codex drafted the scope, architecture, code, test cases, report, and slides from the course requirements. Before submitting, record what the team manually inspected, which tests it actually ran, and what it refined. No team review or meeting is asserted here.

Appendix 1 Meeting register

Date and time | Attendees | Topics discussed | Assignments

Add actual entries from team meetings before submission.